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An Automated Telemedicine System Improves Patient-Reported Well-Being

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Abstract

Background: Control of serum glucose levels is essential for the reduction of complications of diabetes. Telemedicine is one strategy through which serum glucose control can be improved.
Methods: A total of 35 adult, insulin-treated patients with diabetes (type 1 and type 2) were enrolled in the present study (63.0 ± 10 years of age, 63% female) and randomized to telemedicine monitoring (including cordless, remote glucose monitor, and transmitter, $n = 17$), or conventional follow-up ($n = 18$). Metabolic parameters were evaluated, and a quality of life questionnaire was administered both pre- and post-treatment.
Results: Groups were similar at baseline in terms of demographic, quality of life, and metabolic parameters. Significant differences in post-treatment metabolic parameters were not observed, although serum glucose was marginally elevated in the control group compared to the telemedicine group (214 ± 65 mg/dL vs. 171 ± 77 mg/dL, $P = 0.09$). On the other hand, being clinically symptom-free (71% vs. 11%, $P = 0.003$), having no hypoglycemic events (82% vs. 17%, $P = 0.0001$), and having no hyperglycemic events (65% vs. 17%, $P = 0.004$) were all significantly more frequently reported in the telemedicine group compared to the control group. Compared to the control group, the telemedicine group reported experiencing significantly less anxiety, treatment difficulty, depression, disease-associated life complications, and feelings of impotence or ineptitude and significantly greater improvement in personal control over glucose, weight, and overall diabetes.
Conclusions: Though post-treatment metabolic differences were not observed between treatment groups, the telemedicine group reported significantly greater post-treatment experiences of improved quality of life and sense of control over the disease. Thus patient satisfaction can be enhanced through the use of telemedicine.

Introduction

Evidence from large, randomized clinical trials[1,2] as well as from observational studies[3–5] studies indicates that strict metabolic control in both type 1 and type 2 diabetes mellitus (DM) reduces the risk of micro- and macrovascular complications. However, achieving metabolic control is challenged by poor patient compliance.[6]

One way to facilitate metabolic control and, by extension, reduce DM complications is through the use of telemedicine. Studies examining the efficacy of telemedicine in diabetes care indicate that communication of blood glucose monitoring between the patient and the physician appears to be as effective with a modem as a telephone conversation in terms of physician's time spent analyzing data or communicating with the patient, with less error in data transfer.[7]

The Multi-Access Services for the Telematic Management of Diabetes Mellitus project in Italy demonstrated that telemedicine management of diabetes is feasible, well accepted by both physicians and patients, and clinically effective.[8,9]

Access to care can be improved when patients and families receive telehealth care at home rather than in-person care in a clinic or hospital, especially when the telehealth system facilitates self-management that may prevent hospitalization.[10] There is evidence that telemedicine may improve health outcomes.[11]

With improved access to care and health outcomes in mind, the present study was designed to assess the impact of remote monitoring (telemedicine) versus conventional monitoring of patients with diabetes on quality of life, treatment satisfaction, and markers of metabolic control during a 6-month follow-up period.

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Table 1. Demographic and Baseline Metabolic Characteristics by Monitoring Group

	Telemedicine group (n = *17*)	*Control group* (n = *18*)	P *value*
Age (years)	63 ± 10	63 ± 15	0.94
Female sex (%)	10 (59)	12 (67)	0.63
Married (%)	10 (59)	14 (78)	0.23
University education (%)	3 (18)	4 (22)	0.74
HbA1c (%)	8.4 ± 1.4	9.3 ± 1.6	0.09
FBG (mg/dL)	186 ± 82	197 ± 93	0.74
Total cholesterol (mg/dL)	208 ± 51	187 ± 44	0.22
HDL (mg/dL)	47 ± 16	51 ± 18	0.15
LDL (mg/dL)	121 ± 48	100 ± 28	0.49
Triglycerides (mg/dL)	194 ± 133	198 ± 107	0.94
Weight (kg)	78 ± 11	77 ± 12	0.79

HDL, high-density lipoprotein cholesterol; LDL, low-density lipoprotein cholesterol.

Subjects and Methods

Subjects

Study participants were drawn from among insulin-treated patients with type 1 and type 2 diabetes treated at the E. Wolfson Medical Center (Holon, Israel) ambulatory diabetes clinic. All patients were residents of the Holon–Bat Yam–South Tel Aviv–Yaffo area, an urban and suburban region of approximately 500,000 residents. Patients were eligible if they monitored blood glucose with a glucometer more than six times daily at least twice each week and more than three times daily the rest of the week. Patients were adults, 18 years of age and older. All patients signed informed consent, and the study was approved by the Institutional and Ministerial Ethics Committees. Patients were excluded from study participation if they were diagnosed with a concomitant illness including advanced or active cardiovascular disease, neurologic, or psychiatric illness or any terminal illness. Further, patients participating in any other study could not concurrently participate in this study.

Monitoring

Subjects were randomized to telemedicine or conventional monitoring, and patients in both groups were instructed to use their usual glucometer (FreeStyle® [Abbott Diabetes Care, Alameda, CA] or Ascensia® Confirm® [Bayer HealthCare Diabetes Care, Tarrytown, NY]), in the usual manner. In the telemedicine group, glucometers were fitted with a transmitter that automatically remotely transferred data via MedicGate, a receiver that also transmits data to the information server. The system was installed in the home of each participant in the telemedicine group. Information was transferred in real time directly to a computerized medical file. Additionally, results of the glucose measures were simultaneously shown both on the patient's glucometer display and on the diabetes clinic's computer screen. Access to these data was via the internet. For each patient, a range of acceptable glucose values was individually set, and values less than or exceeding that range (from 50 to 280 mg/dL for most patients) triggered an audio alarm at the diabetes clinic computer and generated a text message to the cellular phone of the caregiver, a specialized diabetes nurse (certified by the Ministry of Health to provide medical advice regarding titration of insulin dose, oral medications, etc.). The diabetes nurse specialist then telephoned the patient and provided advice and also notified the diabetes care physician. In nonurgent cases, patients in the telemedicine group were free to contact the diabetes clinic by phone as needed. The telemedicine system used was developed by Telcomed Advanced Industries Ltd., Dublin, Ireland, a subsidiary of the Medic4All Group, Lugano, Switzerland.

Subjects in the conventional monitoring group used their usual glucometers and were free to contact clinic personnel by telephone or visit their outpatient clinic or emergency care unit if they felt glucose values deviated from desired levels. Otherwise, they were scheduled to visit the diabetes clinic every 3 months.

Laboratory analyses

Subjects in both groups provided demographic information and a detailed medical history. Subjects underwent a general physical examination, including blood pressure, pulse rate, weight, and height. Body mass index was calculated. Blood evaluations including glycosylated hemoglobin (HbA1c), fasting plasma glucose, and a lipid profile were undertaken. Subjects completed a quality of life questionnaire that included questions on disease symptoms, hypo- and hyperglycemic event rates, psychological experiences of disease, and disease management. All patients received refresher instructions on use of their glucometer for measuring blood glucose. In the telemedicine group, patients also received instructions on how to utilize the monitoring system. Blood work and the questionnaire were reassessed at the 6-month follow-up point. All laboratory analyses were performed in the Biochemistry Laboratory of the E. Wolfson Medical Center.

Data analysis

Analysis of data was carried out using SPSS version 9.0 (1999) statistical analysis software (SPSS Inc., Chicago, IL). For continuous variables, such as age, blood chemistry variables, and quality of life scores, descriptive statistics were calculated and reported as mean ± standard deviation. Normality of distribution of continuous variables was assessed using the Kolmogorov-Smirnov test (cutoff at $P = 0.01$). Continuous variables were compared by monitoring method using the t test for independent samples or the Mann-Whitney

TABLE 2. BASELINE QUALITY OF LIFE MEASURES BY MONITORING GROUP

	Telemedicine group (n = *17*)	*Control group* (n = *18*)	P *value*
Disease symptoms			
Symptom-free (%)	8 (47)	9 (44)	0.86
No hypoglycemic events (%)	12 (71)	11 (61)	0.55
No hyperglycemic events (%)	3 (18)	4 (22)	0.74
Subjective psychological experience of disease (%)			
No anxiety (%)	2 (12)	2 (11)	0.95
No depression (%)	8 (47)	5 (28)	0.24
No disease-associated difficulty (%)	9 (53)	5 (28)	0.13
No disease-associated complexity (%)	8 (47)	5 (28)	0.24
No feelings of impotence (%)	9 (53)	7 (39)	0.49
No feelings of ineptitude (%)	6 (35)	5 (28)	0.63
Feels control over			
Glucose (%)	4 (24)	5 (28)	0.61
Weight (%)	2 (12)	4 (22)	0.13
Diabetes (%)	1 (6)	2 (11)	0.85
Never requires family assistance with			
Diet (%)	8 (47)	9 (44)	0.86
Medication (%)	14 (82)	14 (76)	0.74
Checking legs/feet for sores (%)	8 (47)	9 (44)	0.86
Exercise (%)	8 (47)	10 (56)	0.62
Glucose monitoring (%)	15 (88)	17 (94)	0.52

U test. General linear modeling (GLM) was used to model post-treatment HbA1c, including monitoring method as a fixed factor and baseline HbA1c (found to marginally differ by monitoring method) as a covariate. Categorical variables such as sex, type of diabetes, and presence of comorbidities were described using frequency distributions and are presented as frequency (percentage). The χ^2 test (exact as needed) was used to assess associations between monitoring method and other categorical variables All tests are two-sided and considered significant at $P < 0.05$.

Results

Demographic and baseline metabolic characteristics of the study population are presented in Table 1. The two groups were similar with the exception of HbA1c, which was marginally higher in the control group (9.3 ± 1.6% vs. 8.4 ± 1.4%, $P = 0.09$). Table 2 presents baseline quality of life measures by monitoring group. Again, the groups were similar in terms of symptoms, hypo- and hyperglycemic event rates, and subjective experience of disease.

Table 3 presents end-of-follow-up metabolic markers. Significant differences were not detected, though fasting blood glucose (FBG) was marginally elevated in the control group compared to the telemedicine group (214 ± 65 mg/dL vs. 171 ± 77 mg/dL, $P = 0.09$). This can be visualized by looking at the difference in change from baseline FBG by monitoring group, as shown in Figure 1. In the telemedicine group, FBG declined by 15 mg/dL, while in the control group, it increased by 18 mg/dL. This difference in change from baseline was not significant ($P = 0.3$) because of large variation in response to treatment for this variable. Because of the marginal difference in HbA1c at baseline, post-treatment HbA1c was adjusted for baseline values. After controlling for baseline differences in HbA1c, post-treatment HbA1c values did not differ by monitoring group.

Post-treatment quality of life measures are presented in Table 4. Significantly more subjects in the telemedicine group reported no disease symptoms than subjects in the control group: 71% versus 11% ($P = 0.003$). Post-treatment, 82% of subjects in the telemedicine group reported having no hypoglycemic events compared to 17% of the subjects in the control group ($P = 0.0001$). Among telemedicine subjects, 65% reported having no hyperglycemic events compared to 17% of the control subjects ($P = 0.004$).

Subjects in the telemedicine group reported experiencing significantly less anxiety, treatment difficulty, depression, disease-associated life complications, and feelings of impotence or ineptitude compared to controls. Disease management was less difficult for subjects in the telemedicine group than among controls, such that 53% of telemedicine subjects compared to 17% of control subjects reported no difficulty with disease management ($P = 0.02$). Consistent with this, subjects in the telemedicine group reported significantly greater improvement in personal control over glucose, weight, and overall diabetes control than subjects in the control group.

Compared to the control group, subjects in the telemedicine group enjoyed a 57% relative increase in the proportion of subjects not requiring family assistance with medications ($P = 0.01$). The need for family assistance with exercise or with examining the feet and legs for sores was significantly reduced in the telemedicine group. By-group differences in requirements for family assistance with diet and glucose monitoring were not detected.

In the telemedicine group only, a questionnaire was completed assessing patient satisfaction with the monitoring system. As shown in Figure 2, the majority of participants in the telemedicine group felt that the system was easy to us and valuable in achieving glucose balance. Further, most expressed that the system improved quality of life and caused changes in disease management and in lifestyle.

TABLE 3. END-OF-FOLLOW-UP METABOLIC MEASURES

	Telemedicine group (n = *17*)	*Control group* (n = *18*)	P *value*
HbA1c (%)	8.5 ± 1.7	9.6 ± 1.9	0.07
Adjusted HbA1c (%)	8.9 ± 1.7	9.2 ± 1.9	0.5
FBG (mg/dL)	171 ± 77	214 ± 65	0.09
Total cholesterol (mg/dL)	177 ± 34	175 ± 37	0.9
HDL (mg/dL)	49 ± 18	50 ± 12	0.37
LDL (mg/dL)	88 ± 22	97 ± 29	0.77
Triglycerides (mg/dL)	182 ± 116	165 ± 99	0.66
Weight (kg)	79 ± 11	77 ± 12	0.78

HDL, high-density lipoprotein cholesterol; LDL, low-density lipoprotein cholesterol.
[a]HbA1c values adjusted for baseline values using GLM, including monitoring group as a fixed factor.

Discussion

Telemonitoring has been shown to facilitate the management of a number of chronic medical conditions, including hypertension,[12] obesity,[13] and heart failure.[14] In individuals with diabetes, strict glucose control is associated with a reduced risk of micro- and macrovascular complications,[1–5] suggesting an important role for enhanced monitoring systems. In a recent systematic review, home telemonitoring in diabetes was associated with reduced HbA1c and complications and improved patient psychological outcomes.[15]

A set of four criteria for assessing the efficacy of a telemedicine program has been proposed[16]: soundness of technology, program efficacy, cost-effectiveness, and practicality.

In the present study, the soundness of technology was demonstrated insofar as no technical "glitches" were encountered and the data were accurately and faithfully collected, incorporated into an electronic database, and communicated to the caregiver, facilitating communication back to the patient.

Program efficacy was suggested by reduced anxiety, depression, and other negative emotions together with an improved sense of control over the disease and reduced need for family assistance with care in the telemedicine group compared to the conventionally monitored participants. These subjective changes occurred in the absence of significant change in metabolic measures. Improved self-reported psychosocial experience despite lack of objectively measurable outcomes has been reported in other populations with diabetes using telemonitoring, including youth with type 1 diabetes[17] and indigent women with gestational diabetes,[18] attributed to the need for concomitant intensive therapy and inadequate system use, respectively. On the other hand, by-group differences in glucose control in the present study were not demonstrated. This may be attributable to small sample size, rendering the study underpowered to detect differences in clinical endpoints such as HbA1c and blood glucose. For example, the increase from baseline HbA1c was more than three times greater in controls than patients in the telemedicine group, but this difference was not significant. A post hoc power calculation indicates that at this sample size (n = 35 total), the study only had 15% power to detect this difference. A total of 32 subjects per group would have been necessary to detect this difference with 80% power. It is conceivable, however, that the between-group difference represents a true zero effect, as has been reported in many studies[19]; indeed, HbA1c and fasting glucose values are not consistent with treatment recommendations in either group. Still, the study was adequately powered to detect by-group differences in quality of life measures, and significant reductions in disease symptoms and hypoglycemic events were observed. This suggests that it is inappropriate to conclude that the use of telemedicine had no impact on clinical outcomes.

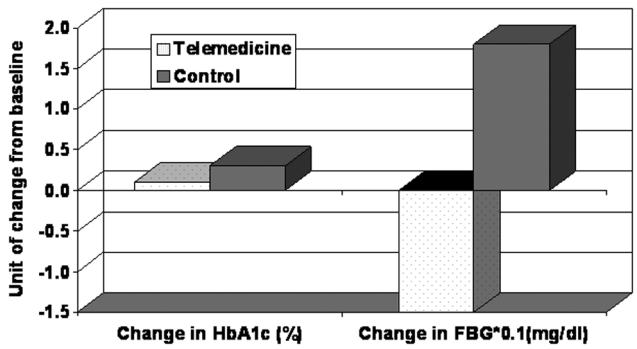

FIG. 1. Change from baseline markers of DM control by group.

It is meaningful that despite the reduction in HbA1c and FBG, hypoglycemic events were also significantly reduced. Furthermore, hyperglycemic events were not more frequently reported in the telemedicine group. These findings are consistent with reduced glucose variability in the telemedicine group and concur with those observed in the 6-year follow-up of the United Kingdom Prospective Diabetes Study, which found that hypoglycemia was an infrequent event among patients following intensive glucose control.[20] It is conceivable that this reduction in glucose excursions could be associated with the reduction in subjective anxiety and could thus ultimately contribute to improved long-term metabolic control.

The present study was designed to examine clinical and quality of life outcomes and not cost-effectiveness. Nevertheless, cost-effectiveness is an important criterion by which a telemedicine system can be assessed. In Israel, health care is nationalized, and the cost of care for a visit to an outpatient diabetes clinic is set at approximately $50 US. This translates to $200 US for four annual outpatient care visits, or $100 US for the study's 6-month follow-up period. This cost

TABLE 4. END-OF-FOLLOW-UP QUALITY OF LIFE MEASURES BY MONITORING GROUP

	Telemedicine group (n = *17*)	*Control group* (n = *18*)	P *value*
Disease symptoms			
Symptom-free (%)	12 (71)	2 (11)	0.003
No hypoglycemic events (%)	14 (82)	3 (17)	0.0001
No hyperglycemic events (%)	11 (65)	3 (17)	0.004
Subjective psychological experience of disease (%)			
No anxiety (%)	9 (53)	3 (17)	0.02
No depression (%)	10 (59)	2 (11)	0.003
No disease-associated difficulty (%)	9 (53)	3 (17)	0.02
No disease-associated complexity (%)	11 (65)	2 (11)	0.001
No feelings of impotence (%)	15 (88)	2 (11)	<0.0001
No feelings of ineptitude (%)	15 (88)	2 (11)	<0.0001
Feels control over			
Glucose (%)	9 (53)	1 (6)	0.002
Weight (%)	9 (53)	2 (11)	0.008
Diabetes (%)	16 (94)	5 (28)	0.0001
Never requires family assistance with			
Diet (%)	8 (47)	6 (33)	0.4
Medication (%)	15 (88)	9 (50)	0.01
Checking legs/feet for sores (%)	15 (88)	5 (28)	0.003
Exercise (%)	13 (77)	4 (22)	0.001
Glucose monitoring (%)	16 (94)	15 (83)	0.32

would be identical between the two groups because the telemedicine group did not forego its routine visits. The difference in cost would be a function of differences in unscheduled visits compared to the cost of initializing the telemedicine system. Because of the nature of Israel's healthcare delivery system, patients can be seen by their healthcare maintenance organization, at an outpatient hospital clinic, through urgent care centers, or at emergency rooms. Other than visits to the outpatient hospital clinic, these visits were not tracked by the present study and so cannot be compared by intervention. It can be stated, however, that none of the patients in the telemedicine group made unscheduled visits to the outpatient diabetes clinic at Wolfson Hospital, while patients in the conventional care group made an average of 2.5 unscheduled visits during the follow-up period. This suggests an addition cost of $125 US in the conventional care group not incurred by the telemedicine group. On the other hand, the cost of setting up the telemedicine system was $150 US per patient, suggesting an additional $25 US during the same time period. However, the extra $25 US is a one-time expense, while it is reasonable to assume that the pattern of unscheduled visits in the conventional care group would continue over time.

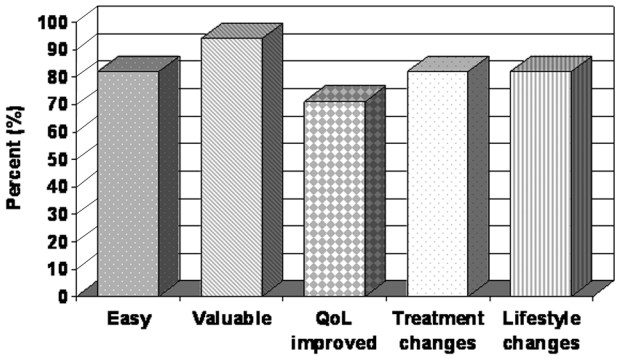

FIG. 2. Satisfaction with telemedicine treatment.

It can also be stated that because Israeli healthcare workers, including diabetes clinic staff, are salaried personnel, their time spent in data evaluation is compensated whether or not the patient is present. This compensation may facilitate the acceptance of telemedicine technology, which has been reported to face reluctance in fee-for-service systems. It cannot be concluded that the telemedicine system was more or less cost-effective that conventional care on the basis of the present report.

Finally, a system must be evaluated on the basis of its practicality. In terms of practicality for the patient, the telemedicine system used in the present study required no new tasks on the part of the patient beyond his or her usual glucose monitoring. Transmission of the data to the server was automated and could be utilized even in homes without personal computers. In terms of practicality for the clinician, the system permitted automated electronic data storage. These data could be easily retrieved and compared over time and incorporated with data acquired at clinic visits.

As with any clinical trial, the degree to which results can be generalized must be considered. All subjects were treated at the E. Wolfson Medical Center, and while the subjects who participated were generally similar to the DM patients treated at the hospital's ambulatory DM clinic, they may not be entirely representative of the Israeli DM population; for example, according to the National Health Survey, females were somewhat over-represented, and ethnic minorities are underrepresented in the study population compared to the national demographic distribution.[21]

A marginal by-group difference in baseline HbA1c was observed in this study population. This difference was controlled for in further analysis, neutralizing its impact. How-

ever, whether this marginal difference in HbA1c reflects other differences as well and, if so, the extent to which these differences might affect outcomes are unknowable. Nevertheless, this point emphasizes the caution required in generalizing results.

In conclusion, though between-group metabolic differences were not detected post-treatment, significantly improved symptoms and a significantly greater proportion of subjects experiencing no hypo- or hyperglycemic events were observed in the telemedicine group. Furthermore, the telemedicine group reported significantly greater post-treatment quality of life and improved sense of control over the disease. These results, together with the reported satisfaction with the monitoring system, indicate that patient care can be enhanced through the use of telemedicine.

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Author Disclosure Statement

There is no conflict of interest for any of the authors.

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